

Lecture 8 – Evaluating estimators and UMVUE

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8.1 Loss and risk

Suppose we have two estimators T_1, T_2 of θ . How do we decide which is better? Statistical decision theory gives us a general framework to do this; in fact, it is an even broader framework for evaluating the performance of all kinds of statistical procedures, including estimation and hypothesis testing.

The decision-theory framework

The basic ingredients of a statistical decision problem are:

- A **statistical model** $\{f_\theta : \theta \in \Theta\}$ for the observations.
- An **action space** $\mathcal{D}$, containing the possible decisions d .
In point estimation, the action d is a proposed value of θ , so the space of possible actions is $\mathcal{D} = \Theta$.
- A **decision rule** $\delta : \mathcal{X}^n \rightarrow \mathcal{D}$, which selects an action $d = \delta(\underline{x})$ after observing the data.
In point estimation, the decision rule $\delta(\underline{X})$ is an estimator of θ , so we write it in the form $\hat{\theta} = \hat{\theta}(\underline{X})$.
- A **loss function** $\mathcal{L}(\theta, d) \geq 0$, measuring the cost of taking action d when the true parameter is θ . We use $\mathcal{L}$ for loss to distinguish it from the likelihood $L_{\underline{x}}(\theta)$.

Example 1 (Loss functions). Two common choices for the loss function are

$$\mathcal{L}(\theta, d) = (d - \theta)^2 \quad (\text{squared-error loss}),$$

and

$$\mathcal{L}(\theta, d) = |d - \theta| \quad (\text{absolute-error loss}).$$

Squared-error loss penalizes large errors more strongly. If overestimating and underestimating have different consequences, one can also choose an asymmetric loss.

The loss $\mathcal{L}(\theta, \hat{\theta}(\underline{X}))$ is random because the sample is random. To evaluate the *procedure*, we average over all samples that could arise under that same θ .

Definition 1 (Risk). The **risk function** of an estimator $\hat{\theta}(\underline{X})$ under loss $\mathcal{L}$ is

$$R(\theta, \hat{\theta}) = \mathbb{E}_\theta[\mathcal{L}(\theta, \hat{\theta}(\underline{X}))].$$

The expectation is over the randomness of $\underline{X}$, under the distribution $f_{\theta, \underline{X}}$. The risk when using the squared-error loss is the most common and gets a special name.

Definition 2 (Mean squared error). The **mean squared error** (MSE) of $\hat{\theta}$ is its risk under squared-error loss:

$$\text{MSE}_{\hat{\theta}}(\theta) = \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2].$$

This should remind you of the variance formula, except that if we were computing the variance, we would have squared $\hat{\theta} - \mathbb{E}_{\theta}[\hat{\theta}]$ rather than $\hat{\theta} - \theta$. It turns out that the MSE decomposes into a sum of two parts: the variance of $\hat{\theta}$, and the *bias* of θ .

Definition 3 (Bias and unbiasedness). The **bias** of $\hat{\theta}$ is

$$B_{\hat{\theta}}(\theta) = \mathbb{E}_{\theta}[\hat{\theta}] - \theta$$

The estimator of θ is **unbiased** if

$$\mathbb{E}_{\theta}[\hat{\theta}] = \theta \quad \text{for every } \theta \in \Theta.$$

Theorem 1 (Bias–variance decomposition). For every $\theta \in \Theta$,

$$\boxed{\text{MSE}_{\hat{\theta}}(\theta) = \text{Var}_{\theta}(\hat{\theta}) + B_{\hat{\theta}}(\theta)^2.}$$

Proof. Fix θ and write $m = \mathbb{E}_{\theta}[\hat{\theta}]$. Add and subtract m , then expand the square:

$$\begin{aligned} \mathbb{E}_{\theta}[(\hat{\theta} - \theta)^2] &= \mathbb{E}_{\theta}[(\hat{\theta} - m) + (m - \theta)]^2 \\ &= \mathbb{E}_{\theta}[(\hat{\theta} - m)^2] + 2(m - \theta)\mathbb{E}_{\theta}[\hat{\theta} - m] + (m - \theta)^2. \end{aligned}$$

The middle term equals zero because $\mathbb{E}_{\theta}[\hat{\theta} - m] = 0$. The first term is $\text{Var}_{\theta}(\hat{\theta})$ and the last term is $B_{\hat{\theta}}(\theta)^2$. $\square$

This tells you the error of an estimator could come from two sources: 1) variability, due to randomness of the data, and 2) error on average: when you average out the possible values of $\hat{\theta}$, you could still be off from θ .

It is generally difficult to have bias and variance small at the same time: there is a trade-off between the two.

Example 2 (Sample mean for a normal). Suppose $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \mathcal{N}(\theta, 1)$, where $\theta \in \mathbb{R}$. Let's find the bias, variance, and MSE of $\hat{\theta} = \bar{X}_n$. We have $\mathbb{E}_{\theta}[\bar{X}_n] = \theta$, so $B_{\bar{X}_n}(\theta) = \theta - \theta = 0$. The estimator is unbiased. For the variance, we have $\text{Var}(\bar{X}_n) = \frac{1}{n} \text{Var}(X_1) = 1/n$. Therefore the MSE is

$$\text{MSE}_{\bar{X}_n}(\theta) = 0^2 + \frac{1}{n} = \frac{1}{n}.$$

In this example, the MSE, bias, and variance all do not depend on θ !

Example 3 (Comparing estimators and bias-variance tradeoff). Consider the same setting as above, but suppose you have a suspicion that θ is small in absolute value. So instead of $\bar{X}_n$, you choose the estimator

$$\hat{\theta}_a = a\bar{X}_n, \quad 0 \leq a \leq 1,$$

where a is fixed in advance. When $a = 1$, we get $\hat{\theta}_1 = \bar{X}_n$, the estimator we considered previously. We compute

$$\mathbb{E}_{\theta}[\hat{\theta}_a] = a\theta, \quad B_{\hat{\theta}_a}(\theta) = (a - 1)\theta, \quad \text{Var}_{\theta}(\hat{\theta}_a) = \frac{a^2}{n},$$

so the bias-variance decomposition gives

$$\text{MSE}_{\hat{\theta}_a}(\theta) = (1 - a)^2\theta^2 + \frac{a^2}{n}.$$

Reducing a lowers the variance but increases the squared bias! This is the **bias–variance tradeoff**. As an extreme example, let's compare the MSE's of $\hat{\theta}_0 = 0$ and $\hat{\theta}_1 = \bar{X}_n$. We get

$$\text{MSE}_{\hat{\theta}_0}(\theta) = \theta^2, \quad \text{MSE}_{\hat{\theta}_1}(\theta) = \frac{1}{n}.$$

We see that $\text{MSE}_{\hat{\theta}_0}(\theta) < \text{MSE}_{\hat{\theta}_1}(\theta)$ when $|\theta| \leq \sqrt{1/n}$. This makes sense: remember that the motivation for using $\hat{\theta}_a$ for some $a < 1$ is that we suspected θ is small. So indeed when θ really is small, $\hat{\theta}_0$ gives a lower MSE than $\hat{\theta}_1$.

Figure 8.1 compares the two MSE's as a function of θ .

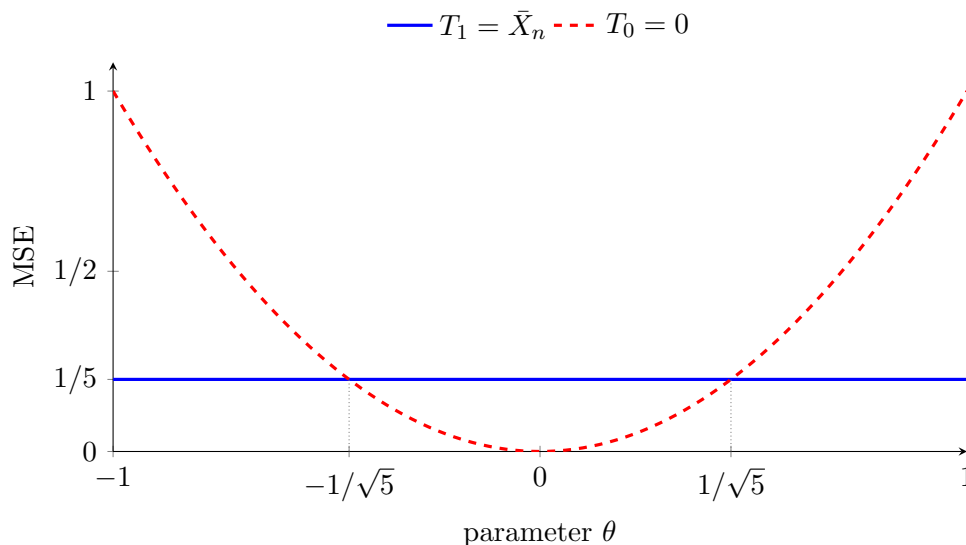


Figure 8.1: Risk functions in Example 2 with $n = 5$. The horizontal curve is $1/5$ and the parabola is θ^2 . Neither estimator has smaller risk for every θ .

The above example showed that for the two estimators we compared, which one has smaller MSE depends on the value of the unknown θ . Ideally, we would like an estimator with smaller MSE *for all values of θ* .

8.2 Uniformly Minimum Variance Unbiased Estimators

Question: How do we define the best estimator?

- In principle, we would like to obtain an estimator that **uniformly** minimizes the risk, i.e., for every possible θ there is no other estimator with lower risk.
- But this is unreasonable. To see why, note that the constant estimator $T(\underline{x}) = c$ has zero risk if $\theta = c$. Thus minimizing the risk uniformly over all possible estimators requires that $\text{MSE}_T(\theta) = 0$ for all θ .
- If we want a notion of **best estimator** we need to restrict the class of possible estimators.

- One option is to restrict to the class of unbiased estimators.

Definition 4. Statistic T^* is the **uniformly minimum-variance unbiased estimator** (UMVUE) for $g(\theta)$ if

$$\begin{aligned}\mathbb{E}_\theta[T^*(\underline{X})] &= g(\theta) && \text{for all } \theta \\ \text{Var}_\theta[T^*(\underline{X})] &\leq \text{Var}_\theta[T(\underline{X})] && \text{for all } \theta \text{ and unbiased estimators } T \text{ for } g(\theta)\end{aligned}$$

We will get into finding the UMVUE in the next few lectures. For now, we discuss some potential issues to be aware of.

1. Sometimes, no unbiased estimator exists. For example, suppose $X_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\theta)$ and the goal is to estimate $g(\theta) = 1/\theta$. Let $U = U(\underline{X})$ be an arbitrary estimator. The sample space is $\{0, 1\}^n$, so the expectation is

$$\mathbb{E}_\theta[U] = \sum_{\underline{x} \in \{0,1\}^n} U(\underline{x}) \theta^{\sum_i x_i} (1-\theta)^{n-\sum_i x_i},$$

which is a polynomial in θ . Therefore, $\mathbb{E}_\theta[U] = 1/\theta$ is impossible.

2. An unbiased estimator might be absurd. For example, suppose $X_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta)$ and the goal is to estimate $g(\theta) = e^{-3\theta}$.

- For $n = 1$, the estimator $T(X) = (-2)^X$ satisfies

$$\mathbb{E}_\theta[T(X)] = \sum_{x=0}^{\infty} (-2)^x \frac{e^{-\theta} \theta^x}{x!} = e^{-\theta} \sum_{x=0}^{\infty} \frac{(-2\theta)^x}{x!} = e^{-\theta} e^{-2\theta} = e^{-3\theta}.$$

and so T is unbiased for $g(\theta)$

- However, T is *negative* if x is odd, even though $g(\theta)$ is strictly positive.
3. Lack of invariance. $\mathbb{E}_\theta[T(\underline{X})] = \theta$ does **not** imply $\mathbb{E}_\theta[g(T(\underline{X}))] = g(\theta)$ in general. So an UMVUE for θ does not immediately give an UMVUE for $g(\theta)$, unlike the MLE.
 4. Some biased estimators might be uniformly better than unbiased ones (in terms of MSE). For example, suppose $X_i \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ and the goal is to estimate $g(\theta) = g(\mu, \sigma^2) = \sigma^2$.

- Consider the estimators

$$S^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \quad \text{and} \quad S_u^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

- It's possible to prove that

$$\mathbb{E}[S^2] = \frac{n-1}{n} \sigma^2 \quad \text{and} \quad \mathbb{E}[S_u^2] = \sigma^2$$

and so S^2 is *not* unbiased.

- It can also be shown that S^2 is uniformly better in the sense that

$$\text{MSE}_{S^2}(\theta) < \text{MSE}_{S_u^2}(\theta), \quad \forall \theta$$